PARAMVEER SINGH

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EDUCATION

Thapar Institute of Engineering & Technology, Patiala **Aug 2024 – Present**
Bachelor of Engineering in Computer Engineering **CGPA: 9.49/10.00**

Meritorious School, Jagraon **2021 – 2024**
Class XII: 98.3% Class X: 94.7%

EXPERIENCE

Research Intern, Experiential Learning Centre (ELC), Thapar Institute of Engineering & Technology **2026**
Vision-Based Powder Flow Process Parameter Optimization for Directed Energy Deposition of Waste Metallic Particles
Python, OpenCV, Scikit-Learn, Statsmodels **GitHub**

- Built an end-to-end computer-vision pipeline extracting 15 physically motivated features (stream geometry, density, symmetry, dynamics) from 36 directed energy deposition (DED) powder-flow experiments, using mass-weighted edge detection, dense optical flow, and FFT-based periodicity analysis.
- Designed a multi-criteria decision-making (MCDM) framework combining PCA-filtered TOPSIS, AHP, entropy weighting, and MOORA to rank process conditions; the same three conditions ranked in the top 5 under all six methods (pairwise Spearman $\rho \geq 0.81$), and the top condition stayed in the top 5 in 100% of 500 Monte Carlo weight perturbations.
- Diagnosed a measurement confound in which video features tracked powder feed rate (ρ up to 0.98) and corrected it with OLS regression, reducing the correlation to $|\rho| < 0.25$.
- Fitted a quadratic response surface ($R^2 = 0.82$, $p < 0.001$) identifying carrier gas as the dominant parameter, and confirmed the top condition in each of three spot-diameter zones through DED printing trials; manuscript in preparation.

PROJECTS

Aircraft Engine Predictive Maintenance **GitHub**
PyTorch, Scikit-Learn, Pandas, NumPy

- Built a leakage-free Remaining Useful Life (RUL) pipeline on NASA C-MAPSS (engine-level train/validation split, train-only scaling, piecewise-linear RUL target) and trained LSTM and Transformer models with early stopping over 3 seeds, reaching test RMSE 13.4 and 13.1 on FD001, a 68% reduction over a mean-predictor baseline.
- Benchmarked against Ridge and Random Forest using RMSE, R^2, the asymmetric NASA PHM08 score and bootstrap confidence intervals; a feature-engineered Random Forest (RMSE 12.8) performed comparably to the deep models, and error analysis showed near-failure predictions within 3.9 cycles RMSE.
- Showed via ablation that positional encoding cuts Transformer RMSE from 15.9 to 12.9 and that a 30-cycle window outperforms 20 (15.7); trained an autoencoder on healthy engine states with a threshold calibrated on held-out engines, detecting near-failure windows at AUROC 0.76.

TIES – Campus Mobility Platform **GitHub**
React.js, Vite, Tailwind CSS, React Router, JavaScript

- Built the frontend of a campus carpooling app in React and Tailwind CSS: responsive layouts, dark mode, and the full ride flow, from posting and joining rides to driver approvals via notifications.
- Implemented Google Sign-In and a fetch wrapper that silently refreshes expired JWTs through one shared request, so parallel API calls never trigger duplicate refreshes.

SKILLS

Languages: C++, Python, JavaScript, SQL
ML/DL: PyTorch, Scikit-Learn, NumPy, Pandas, Matplotlib
Frameworks: React.js, Node.js, Express.js, MongoDB
Core CS: Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks
Tools: Git, GitHub, Docker, VS Code, Postman, Jupyter Notebook

ACHIEVEMENTS

- 2× Full Merit Scholarship Recipient, Thapar Institute of Engineering & Technology.
- 2× Gold Medal – InterTech Basketball Tournament (IIT Ropar,PEC Chandigarh).
- Silver Medal – Inter-University Basketball Tournament (IIT Mandi).
- 2× Gold Medalist – URJA Sports Fest.

LEADERSHIP & RESPONSIBILITY

- **Basketball Coordinator, URJA Sports Fest** – Coordinated participants and managed execution of large-scale sporting events.
- **Technical Head, VIRSA Society** – Led technical planning and execution for cultural events, coordinating student participation initiatives.
- Represented TIET in inter-university basketball tournaments and contributed to multiple medal-winning performances.